

Adaptive Position-Guided Multi-Armed Bandit for Beam Refinement in mmWave RIS

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Abstract—Reconfigurable Intelligent Surfaces (RISs) are envisioned as key enablers for future Millimeter Wave (mmWave) networks by providing programmable radio environments with low-cost and energy-efficient beamforming. Fast and accurate beam alignment remains challenging due to the huge training overhead of exhaustive codebook search and the sensitivity of Position Information (PI)-based approaches to localization errors. An *adaptive PI-guided multi-level Multi-Armed Bandit (MAB)* approach is proposed for RIS-assisted mmWave communication. The algorithm refines beam selection by combining PI-based initialization with online MAB learning and hierarchical codebook updates, ensuring robustness against position errors. Simulation results show that the proposed method achieves near-optimal SNR and energy efficiency while significantly reducing training overhead. It outperforms conventional PI- and MAB-based baselines and surpasses a dense PI-based sweeping method that uses 450 codewords, achieving similar performance with only 30 codewords. This corresponds to a $15\times$ reduction in beam evaluations. The approach is well aligned with digital twin frameworks for RIS-assisted beam management, where virtual models provide priors and online adaptation enables efficient real-time operation.

Index Terms—beamforming, mmWave, RIS, digital twin

I. INTRODUCTION

The emergence of Reconfigurable Intelligent Surface (RIS) has sparked a paradigm shift in the design of wireless networks operating at high-frequency bands such as mmWave and Tera-Hertz (THz). By enabling programmable radio environments, RISs provide cost-effective and energy-efficient alternatives to conventional relaying and massive MIMO solutions. The network performance gains offered by RIS critically depend on effective beam alignment, which in turn requires configuring a large number of reflecting elements. Effective beam alignment is often hindered by the substantial training overhead and complexity associated with acquiring accurate Channel State Information (CSI), especially in dynamic or dense deployments [1].

To reduce the feedback overhead, codebook-based beamforming strategies have been widely deployed and are already part of practical 5G NR systems, where predefined beam codebooks are used for initial access and beam management. These approaches serve as practical alternatives to full CSI-based optimization. Existing methods typically rely on exhaustive or hierarchical searches over such codebooks to identify the best RIS configuration, such as the set of phase shifts applied across the RIS elements. Although an exhaustive search guarantees optimality, its complexity scales unfavorably with the size of

the codebook [2]. Hierarchical beam training, on the other hand, achieves a reasonable trade-off by progressively refining the beam direction using a multi-level codebook structure [3]. Nevertheless, in RIS-assisted scenarios, additional challenges arise due to the fully passive nature of RIS, which usually precludes conventional CSI feedback from the surface. Furthermore, at high-frequency bands where very large antenna arrays are employed, CSI acquisition and feedback reporting become excessively costly in terms of signaling overhead. As a result, conventional codebook-based strategies still suffer from significant search latency and scalability issues, particularly under uncertainty or rapid channel variations [4].

To overcome these limitations, Position Information (PI) has been recently incorporated into RIS beamforming schemes, where location estimates of User Equipment (UE) guide the selection of candidate beams from the codebook [5], [6]. While such approaches dramatically reduce training overhead, their effectiveness diminishes under location errors and environmental uncertainty. Moreover, most PI-based methods assume offline-trained codebooks or deterministic mappings between position and beam index, limiting adaptability in practice. In this context, Digital Twin (DT) frameworks are emerging as a promising tool to support the design, testing, and optimization of such beam selection strategies in virtual environments. For example, a functional DT can generate diverse channel and position realizations, enabling efficient development and validation of robust algorithms under controlled conditions [7]. Common approaches for integrating DTs into codebook-based beamforming include building twins based on realistic channel models with 5G-NR feedback emulation, where deep generative models are trained to enable robust beamforming under limited feedback [8].

In this paper, we propose a *PI-guided Multi-Armed Bandit (MAB) framework* for codebook-based RIS beamforming, designed to be compatible with DT systems. The method uses lightweight real-time feedback from the UE to adaptively refine beam selection under location uncertainty and channel variability. The initial beam search region is determined probabilistically based on the estimated user direction, which could also be provided or augmented by a digital twin using environment-aware position estimates or statistical beam likelihoods. While our evaluation is conducted in a simulated setup, the framework is well suited for future integration into DT-enabled beamforming pipelines.

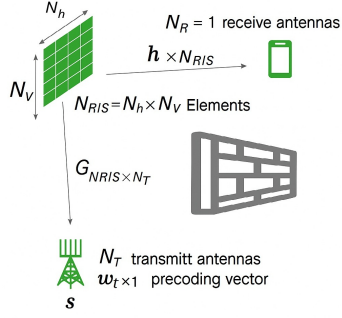


Fig. 1: Illustration of the RIS-assisted mmWave system

II. SYSTEM MODEL

We consider a point-to-point downlink mmWave system where a Base Station (BS) equipped with a uniform planar array (UPA) of N_T antennas communicates with a single-antenna UE via a RIS, as illustrated in Figure 1. The RIS is composed of passive reflecting elements $N_{\text{RIS}} = N_h \times N_v$, arranged in a UPA with N_h horizontal and N_v vertical elements. Both the BS and RIS are deployed at fixed and known locations. This paper considers the case where the direct path between BS and UE is assumed to be completely blocked due to obstacles. Thus, all communication occurs through the RIS.

A. Transmission Model

Let $s \in \mathbb{C}$ denote the transmitted symbol, and let $\mathbf{w} \in \mathbb{C}^{N_T \times 1}$ be the BS precoding vector. The RIS applies a diagonal phase shift matrix $\Phi = \text{diag}(e^{j\theta_1}, \dots, e^{j\theta_{N_{\text{RIS}}}})$, where $\theta_i \in [0, 2\pi)$ denotes the phase shift at element i . The received signal at the UE is

$$y = \frac{1}{\sqrt{L_R}} \mathbf{h}^H \Phi \mathbf{G} \mathbf{w} s + n, \quad (1)$$

where $\mathbf{G} \in \mathbb{C}^{N_{\text{RIS}} \times N_T}$ is the BS-RIS channel, $\mathbf{h} \in \mathbb{C}^{N_{\text{RIS}} \times 1}$ is the RIS-UE channel, and $n \sim \mathcal{CN}(0, \sigma^2)$ is additive noise.

Both $\mathbf{G}$ and $\mathbf{h}$ are modeled using the 3GPP Clustered Delay Line (CDL) model to reflect realistic small-scale multipath fading at mmWave frequencies [9]. The CDL captures parameters such as the number of clusters, delay spread, and angular spread, but it does not include large-scale path loss. Therefore, the total path loss for the RIS-assisted link, denoted L_R , is applied externally and calculated according to the ETSI far-field model [10].

$$L_R = \left(\frac{4\pi d_T d_R}{N_{\text{RIS}} d_h d_v A} \right)^2, \quad (2)$$

where d_T and d_R are the distances from BS to RIS and from RIS to UE, respectively, d_h and d_v are the dimensions of RIS elements along horizontal and vertical directions, and A is the reflection amplitude coefficient.

B. Problem Statement

Our objective is to maximize the SNR at the UE by jointly optimizing the BS precoder $\mathbf{w}$ and the RIS phase shifts $\{\theta_1, \dots, \theta_{N_{\text{RIS}}}\}$. In practice, the absence of CSI feedback

from the RIS and the uncertainty in user location prevent full CSI-based optimization. To address these constraints, we adopt a codebook-based beamforming strategy, where a finite RIS codebook $\mathcal{C} = \{\Phi_1, \dots, \Phi_Q\}$ containing Q possible configurations is considered. The optimal beam configuration is then obtained as

$$q^* = \arg \max_{q \in \{1, \dots, Q\}} \frac{1}{L_R} |\mathbf{h}^H \Phi_q \mathbf{G} \mathbf{w}|^2, \quad (3)$$

For a fixed $\mathbf{w}$, the problem reduces to selecting the optimal Φ_q from the codebook $\mathcal{C}$. In the presence of user location uncertainty, the SNR can be defined in expectation over the position error distribution.

III. CODEBOOK BASED SOLUTIONS

In this section, we review recent codebook-based beamforming strategies, including hierarchical [3] and PI-guided [5] search methods, that are relevant to our proposed framework. All methods are evaluated under the common system model defined in Section II, using a CDL-based RIS-assisted simulation environment.

A. Hierarchical Beam Search (HBS)

The objective of HBS is to identify the RIS configuration that maximizes the SNR at the UE while significantly reducing beam-training overhead versus exhaustive search. The method proceeds through L refinement levels indexed by $l = 1, \dots, L$.

At the first level ($l = 1$), the RIS azimuth and elevation grids, $\mathcal{A}_{\text{RIS}}^{(1)}$ and $\mathcal{E}_{\text{RIS}}^{(1)}$, cover the angular domain of interest with coarse resolution. The BS azimuth and elevation grids, $\mathcal{A}_{\text{BS}}$ and $\mathcal{E}_{\text{BS}}$, are defined at the beginning of the search.

For each candidate RIS beam $(\phi, \theta) \in \mathcal{A}_{\text{RIS}}^{(l)} \times \mathcal{E}_{\text{RIS}}^{(l)}$, the corresponding codeword (phase-shift matrix) is constructed as:

$$\Phi(\phi, \theta) = \text{diag}\{e^{j\theta_{m,n}}\}_{m=1, n=1}^{N_h, N_v}, \quad (4)$$

$$\theta_{m,n} = \text{mod}_{2\pi} \left(-\frac{2\pi}{\lambda} [(m-1)\Delta_x, (n-1)\Delta_y, 0] \cdot \mathbf{u}(\phi, \theta) \right)$$

where $\mathbf{u}(\phi, \theta) = [\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta]$ is the unit direction vector and λ is the carrier wavelength.

For each level $l > 1$, the system evaluates all combinations of BS and RIS beams in the current grids and records the configuration with the maximum SNR. In this work, only the *best beam tuple* $(\phi_{\text{BS}}^{*(l)}, \theta_{\text{BS}}^{*(l)}, \phi_{\text{RIS}}^{*(l)}, \theta_{\text{RIS}}^{*(l)})$ is retained for refinement. The RIS grids are then narrowed and re-sampled around $(\phi_{\text{RIS}}^{*(l)}, \theta_{\text{RIS}}^{*(l)})$. Let M_ϕ and M_θ be the numbers of azimuth and elevation samples per level, and let the level- l steps be $\Delta_\phi^{(l)}$ and $\Delta_\theta^{(l)}$. We uniformly sample a *symmetric* window centered at $(\phi_{\text{RIS}}^{*(l)}, \theta_{\text{RIS}}^{*(l)})$ with *total* angular spans $W_\phi^{(l)}$ and $W_\theta^{(l)}$, so $\Delta_\phi^{(l)} = \frac{W_\phi^{(l)}}{M_\phi - 1}$, and $\Delta_\theta^{(l)} = \frac{W_\theta^{(l)}}{M_\theta - 1}$.

At each level, the window is shrunk by a factor $\rho \in (0, 1)$ as given in Alg. 1, $W_\phi^{(l+1)} = \rho W_\phi^{(l)}$, $W_\theta^{(l+1)} = \rho W_\theta^{(l)}$ which

Alg. 1 Hierarchical Beam Search (HBS)

Inputs: Initial RIS grids $\mathcal{A}_{\text{RIS}}^{(1)}, \mathcal{E}_{\text{RIS}}^{(1)}$; fixed BS grids $\mathcal{A}_{\text{BS}}, \mathcal{E}_{\text{BS}}$; levels L ; samples per level M_ϕ, M_θ ; shrink factor $\rho \in (0, 1)$
Output: $(\phi_{\text{RIS}}^*, \theta_{\text{RIS}}^*)$
Initialize window widths: $W_\phi^{(1)} = \max \mathcal{A}_{\text{RIS}}^{(1)} - \min \mathcal{A}_{\text{RIS}}^{(1)}$, $W_\theta^{(1)} = \max \mathcal{E}_{\text{RIS}}^{(1)} - \min \mathcal{E}_{\text{RIS}}^{(1)}$
Initialize global best $(\gamma^*, \phi_{\text{RIS}}^*, \theta_{\text{RIS}}^*)$
for level $l = 1$ to L **do**
 for each $(\phi_{\text{RIS}}, \theta_{\text{RIS}}) \in \mathcal{A}_{\text{RIS}}^{(l)} \times \mathcal{E}_{\text{RIS}}^{(l)}$ **do**
 for each $(\phi_{\text{BS}}, \theta_{\text{BS}}) \in \mathcal{A}_{\text{BS}} \times \mathcal{E}_{\text{BS}}$ **do**
 Configure BS precoder with $(\phi_{\text{BS}}, \theta_{\text{BS}})$
 Configure RIS with $(\phi_{\text{RIS}}, \theta_{\text{RIS}})$
 $\gamma \leftarrow \text{SNR}(\phi_{\text{BS}}, \theta_{\text{BS}}, \phi_{\text{RIS}}, \theta_{\text{RIS}})$
 if $\gamma > \gamma^{(l)}$ **then**
 $(\gamma^{(l)}, \phi_{\text{RIS}}^{*(l)}, \theta_{\text{RIS}}^{*(l)}) = (\gamma, \phi_{\text{RIS}}, \theta_{\text{RIS}})$
 end if
 if $\gamma > \gamma^*$ **then**
 $(\gamma^*, \phi_{\text{RIS}}^*, \theta_{\text{RIS}}^*) = (\gamma, \phi_{\text{RIS}}, \theta_{\text{RIS}})$
 end if
 end for
 end for
 if $l < L$ **then**
 Refine the RIS codebook by narrowing the angular window using Equations (5) and (6) and uniformly resampling around the best RIS direction:
 $W_\phi^{(l+1)} = \rho W_\phi^{(l)}, W_\theta^{(l+1)} = \rho W_\theta^{(l)}$
 $\Delta_\phi^{(l+1)} = \frac{W_\phi^{(l+1)}}{M_\phi - 1}, \Delta_\theta^{(l+1)} = \frac{W_\theta^{(l+1)}}{M_\theta - 1}$
 end if
end for
return $(\phi_{\text{RIS}}^*, \theta_{\text{RIS}}^*)$

implies $\Delta_\phi^{(l+1)} = \rho \Delta_\phi^{(l)}, \Delta_\theta^{(l+1)} = \rho \Delta_\theta^{(l)}$. Level- $(l+1)$ grids can be expressed as follows.

$$\mathcal{A}_{\text{RIS}}^{(l+1)} = \left\{ \phi_{\text{RIS}}^{*(l)} - \frac{W_\phi^{(l+1)}}{2} + k \Delta_\phi^{(l+1)} \mid k = 0, \dots, M_\phi - 1 \right\}, \quad (5)$$

$$\mathcal{E}_{\text{RIS}}^{(l+1)} = \left\{ \theta_{\text{RIS}}^{*(l)} - \frac{W_\theta^{(l+1)}}{2} + k \Delta_\theta^{(l+1)} \mid k = 0, \dots, M_\theta - 1 \right\}, \quad (6)$$

with angles clipped to the FoV if needed. The symmetric index set is defined as $\mathcal{I}_M = \{-\lfloor \frac{M}{2} \rfloor, \dots, \lfloor \frac{M-1}{2} \rfloor\}$, which spans M integer values centered around zero. After all L levels, the RIS azimuth and elevation that achieve the highest SNR are selected. The overall procedure is summarized in Alg. 1 and the time complexity of the hierarchical beam search is $\mathcal{O}(L \cdot M_\phi M_\theta \cdot N_{\text{BS}})$, where L is the number of refinement levels, M_ϕ and M_θ are the number of RIS azimuth and elevation samples per level, and $N_{\text{BS}} = |\mathcal{A}_{\text{BS}}| \cdot |\mathcal{E}_{\text{BS}}|$ is the number of beam directions tested at the base station.

B. PI-Guided Codebook-Based RIS Beamforming

The PI-guided RIS beamforming method aims to reduce beam training overhead by restricting the RIS codeword search to directions that are geometrically consistent with the estimated user position. Specifically, instead of evaluating the full RIS codebook $\mathcal{C} = \{\Phi_1, \dots, \Phi_Q\}$ exhaustively, the method selects a reduced subset $\mathcal{C}_{\text{sub}} \subset \mathcal{C}$ containing only the K codewords that have a direction closest to the estimated user direction.

Alg. 2 PI-Guided Codebook Selection

Input: RIS codebook $\mathcal{C} = \{\Phi_1, \dots, \Phi_Q\}$, true position $\mathbf{r}_{\text{UE}}$, position noise variance σ_{pos}^2 , angular subset size K , N_{trials}
Output: Best codeword index q^* per trial
for $t = 1$ to N_{trials} **do**
 $\hat{\mathbf{r}}_{\text{UE}}^{(t)} = \mathbf{r}_{\text{UE}} + \mathbf{e}^{(t)}, \mathbf{e}^{(t)} \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{pos}}^2 \mathbf{I})$
 Compute estimated angles $(\hat{\theta}_{\text{az}}, \hat{\theta}_{\text{el}})$ from RIS to UE
 Compute angular distance from each codeword in $\mathcal{C}$ to $(\hat{\theta}_{\text{az}}, \hat{\theta}_{\text{el}})$
 Select subset $\mathcal{C}_{\text{sub}}$ of K nearest codewords
 for each $\Phi_q \in \mathcal{C}_{\text{sub}}$ **do**
 Configure RIS with Φ_q
 Apply MRT-based BS precoder
 Measure SNR $\gamma_q^{(t)}$
 end for
 Select $q^* = \arg \max \gamma_q^{(t)}$
end for
return $\{q^*\}_{t=1}^{N_{\text{trials}}}$ and corresponding SNRs

To model location uncertainty, the actual user position $\mathbf{r}_{\text{UE}}$ is perturbed with Gaussian noise $\mathbf{e}^{(t)} \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{pos}}^2 \mathbf{I})$ to obtain the estimated position $\hat{\mathbf{r}}_{\text{UE}}$. From the estimated user location, the azimuth and elevation angles $(\hat{\theta}_{\text{az}}, \hat{\theta}_{\text{el}})$ from the RIS to the UE are computed. Each RIS codeword is associated with a discrete azimuth and elevation pair. An Euclidean angular distance metric is used to identify the K nearest codewords to the PI direction.

The execution of the PI-guided beamforming algorithm consists of multiple independent Monte Carlo trials, where each trial simulates an estimated UE position drawn from a Gaussian distribution centered at a fixed true location. This setup models location uncertainty using a predetermined standard deviation. During each trial, the angularly filtered codebook $\mathcal{C}_{\text{sub}}$ is swept exhaustively. To optimize the RIS phase shifts, we adopt the suboptimal iterative method proposed in [11], which alternates between updating the precoding vector using Maximum Ratio Transmission (MRT) and refining the RIS phase shifts to align the composite channel phase, repeating the process until convergence. The Signal to Noise Ratio (SNR) is measured for all K codewords, and the codeword yielding the highest SNR is selected. Compared to exhaustive or hierarchical beam search, the PI-guided approach allows the number of beams to be significantly reduced by selecting a smaller angular subset size K , while still maintaining high accuracy through geometric alignment with the estimated user direction. The pseudo-code is provided in Alg. 2 and its time complexity can be expressed as $\mathcal{O}(N_{\text{trials}} \cdot Q)$, where $Q = |\mathcal{C}|$ is the total number of codewords and N_{trials} is the number of perturbed UE positions sampled.

IV. PROPOSED ADAPTIVE PI-GUIDED MULTI-LEVEL MAB BEAMFORMING

This section introduces a MAB-based algorithm for RIS codeword selection. Both the baseline and the proposed methods follow the same core MAB strategy, but differ in how they initialize and refine the RIS codebook.

Alg. 3 MAB-Based RIS Codeword Selection

Input: $\mathcal{C}$, $\mathbf{r}_{\text{UE}}$, σ_{pos}^2 , max trials M
Output: Selected RIS codeword index q^*
Perturb position: $\hat{\mathbf{r}}_{\text{UE}} = \mathbf{r}_{\text{UE}} + \mathbf{e}$
Compute path loss L_{R}
Initialize MAB counts $C_q = 0$, rewards $R_q = 0$ for all q
for $k = 1$ to M **do**
 Select q_k via UCB rule
 Apply Φ_{q_k} and measure SNR γ_{q_k}
 Update $R_{q_k} = R_{q_k} + \gamma_{q_k}$, $C_{q_k} = C_{q_k} + 1$
end for
 $q^* = \arg \max_q R_q / C_q$
return q^*

A. MAB-Based RIS Codebook Selection

The MAB-based RIS codebook selection method aims to maximize the received SNR at the UE by adaptively selecting the RIS configuration from a predefined codebook $\mathcal{C} = \{\Phi_1, \dots, \Phi_Q\}$, where each codeword Φ_q corresponds to a specific RIS phase shift matrix. The goal is to identify the index of the best codeword, q^* that maximizes the effective SNR (cf., equation (3)). At each trial, the actual UE position $\mathbf{r}_{\text{UE}}$ is perturbed to model location uncertainty as follow.

$$\hat{\mathbf{r}}_{\text{UE}} = \mathbf{r}_{\text{UE}} + \mathbf{e}, \quad \mathbf{e} \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{pos}}^2 \mathbf{I}). \quad (7)$$

The player in the MAB problem applies the Upper Confidence Bound (UCB) rule to select the next RIS codeword.

$$q_k = \arg \max_q \left(\frac{R_q}{C_q} + \sqrt{\alpha \frac{\log(T+1)}{C_q}} \right), \quad (8)$$

where R_q is the cumulative reward, which is SNR, C_q is the selection count for the codeword q , $T = \sum_q C_q$ is the total number of selections, and α controls the exploration. At each selection, the reward is updated:

$$R_q = R_q + \gamma_q, \quad C_q = C_q + 1, \quad (9)$$

where γ_q is the measured SNR for codeword q . After the specified number of trials, the codeword is selected as expressed in Equation (3). Alg. 3 summarizes the baseline MAB method and its time complexity is $\mathcal{O}(M \cdot Q)$.

B. Proposed MAB-Based RIS Codebook Selection

The proposed algorithm extends the MAB-based codeword selection introduced in the previous section. While the baseline method adaptively explores a fixed codebook $\mathcal{C} = \{\Phi_1, \dots, \Phi_Q\}$ using the UCB rule in Equation (8) and updates rewards as in Equation (9), the proposed approach improves performance by incorporating *PI*-based codebook initialization and *multi-level refinement*. These additions enable more focused exploration of the angular domain and faster convergence with fewer beam measurements. The objective of the proposed algorithm is to identify the best RIS codeword that maximizes the received SNR at the UE as expressed in Equation (3).

1) *PI-Based Angular Region Selection*: At each trial, the UE position is perturbed to model uncertainty, as expressed with Equation (7).

From $\hat{\mathbf{r}}_{\text{UE}}$, the estimated RIS angles $(\hat{\phi}_{\text{RIS}}, \hat{\theta}_{\text{RIS}})$ are derived geometrically. A *PI-based codebook* is then constructed by restricting to the angular neighborhood of $(\hat{\phi}_{\text{RIS}}, \hat{\theta}_{\text{RIS}})$:

$$\mathcal{C}_{\text{PI}} = \{\Phi_q \in \mathcal{C} \mid \hat{\phi}_{\text{RIS}} - \Delta_\phi \leq \phi_q \leq \hat{\phi}_{\text{RIS}} + \Delta_\phi, \quad \hat{\theta}_{\text{RIS}} - \Delta_\theta \leq \theta_q \leq \hat{\theta}_{\text{RIS}} + \Delta_\theta\}, \quad (10)$$

where $\Delta_\phi, \Delta_\theta$ are angular thresholds proportional to the standard deviation of the location error. The initial codebook is then defined as $\mathcal{C}_{\text{init}} = \mathcal{C}_{\text{PI}} \cup \mathcal{C}_{\text{coarse}}$, where $\mathcal{C}_{\text{coarse}}$ is a subset of coarsely spaced beams selected from outside the PI region to ensure exploration beyond the localized angular neighborhood and avoid local trapping.

2) *MAB Learning Phase*: Within $\mathcal{C}_{\text{init}}$, each codeword q is assigned a prior weight based on angular proximity to $(\hat{\phi}_{\text{RIS}}, \hat{\theta}_{\text{RIS}})$ as follows.

$$w_q^{(0)} \propto \exp \left(-\frac{(\phi_q - \hat{\phi}_{\text{RIS}})^2}{2\sigma_\phi^2} - \frac{(\theta_q - \hat{\theta}_{\text{RIS}})^2}{2\sigma_\theta^2} \right), \quad (11)$$

where $(\sigma_\phi, \sigma_\theta)$ parameterize the location error distribution. Both parameters approximate the angular standard deviations resulting from the location uncertainty σ_{pos} in Equation (7). Although the transformation from position to angles is non-linear, these parameters can be treated as design variables that capture the effective spread of the angular error distribution, which is approximately Gaussian for small σ_{pos} .

The MAB learning process is repeated for a total of M iterations, where each iteration selects one RIS codeword using the UCB rule in Equation (8). It updates the statistics (R_q, C_q) according to Equation (9).

3) *Multi-Level Codebook Update*: To accelerate convergence, the codebook is periodically refined around the best-performing beam $(\phi_{\text{best}}, \theta_{\text{best}})$ found so far as shown below.

$$\mathcal{C}_{\text{update}} = \{\Phi_q \in \mathcal{C} \mid \phi_{\text{best}} - \Delta_\phi \leq \phi_q \leq \phi_{\text{best}} + \Delta_\phi, \quad \theta_{\text{best}} - \Delta_\theta \leq \theta_q \leq \theta_{\text{best}} + \Delta_\theta\}, \quad (12)$$

where $(\Delta_\phi, \Delta_\theta)$ are refinement steps that shrink across levels. In practice, refinement is triggered after a fixed number of trials, such as at $k = [M/2]$ and $k = [3M/4]$, to balance exploration and refinement. The pseudo-code of the proposed algorithm is given in Alg. 4 with a time complexity of $\mathcal{O}(Q + M \cdot Q_{\text{max}})$, where $Q_{\text{max}} < Q$ reflects the reduced codebook size after PI filtering and adaptive refinement.

V. PERFORMANCE RESULTS

In this section, we evaluate the performance of the proposed adaptive PI-guided multi-level MAB beamforming algorithm in a RIS-assisted mmWave communication system. The considered scenario consists of a BS equipped with a uniform planar array, a passive RIS deployed at a fixed position, and a single-antenna UE. The system parameters adopted in the simulations are summarized in Table I.

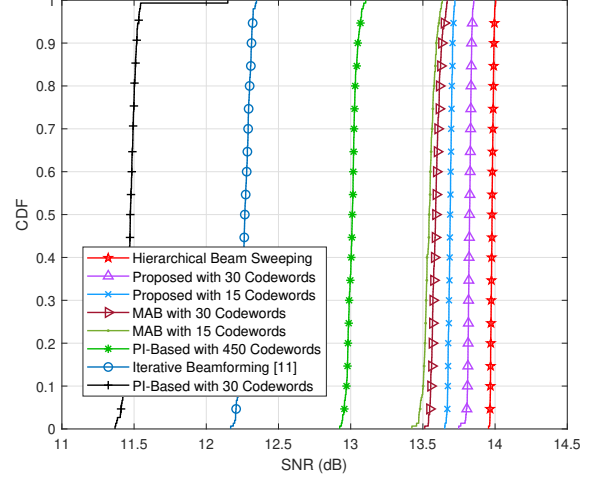
Alg. 4 Adaptive PI-Guided Multi-Level MAB Beamforming

Input: Full codebook $\mathcal{C}$, UE position $\mathbf{r}_{\text{UE}}$, error variance σ_{pos}^2 , angular spreads $(\sigma_\phi, \sigma_\theta)$, max trials M
Output: Best RIS codeword index q^*
Sample perturbed position $\hat{\mathbf{r}}_{\text{UE}}$
Compute estimated RIS angles $(\hat{\phi}_{\text{RIS}}, \hat{\theta}_{\text{RIS}})$ Form $\mathcal{C}_{\text{PI}}$ using Eq. (11), then define $\mathcal{C}_{\text{init}} = \mathcal{C}_{\text{PI}} \cup \mathcal{C}_{\text{coarse}}$
Initialize MAB statistics with prior weights
for $k = 1$ to M **do**
 Select codeword q_k by UCB rule as in Equation (8)
 Apply Φ_{q_k} , measure SNR γ_{q_k}
 Update $R_{q_k} = R_{q_k} + \gamma_{q_k}$, $C_{q_k} = C_{q_k} + 1$
 if $k = \lfloor M/2 \rfloor$ or $k = \lfloor 3M/4 \rfloor$ (refinement step) **then**
 Refine codebook around $(\phi_{\text{best}}, \theta_{\text{best}})$ using Equation (12)
 Reinitialize MAB statistics over updated codebook
 end if
end for
 $q^* = \arg \max_q R_q / C_q$
return q^*

TABLE I: Simulation Parameters

Parameter	Value
Carrier frequency f_c	28 GHz
Subcarrier spacing	120 kHz
BS antenna array size	$4 \times 2 \times 2$
RIS array size	$8 \times 4 \times 2$
d_h and d_v	$\lambda/5$
Reflection amplitude coefficient for RIS elements	0.8
Number of RIS elements	64
Tx power	20 dBm
Noise figure	7 dB
Thermal noise density	-174 dBm/Hz
Bandwidth	72 MHz
BS location	$(-105, 0, 0)$ m
RIS location	$(0, 0, 0)$ m
UE location	$(0, 7, 0)$ m

Performance comparison of the proposed algorithm is illustrated in Figure 2. The results are obtained at the actual UE location, with no position error, $\sigma_{\text{pos}}^2 = 0$. The upper bound is achieved by hierarchical beam sweeping with a 90th-percentile SNR of 13.99 dB. The proposed algorithm achieves 13.83 dB using 30 codewords and 13.70 dB using only 15 codewords, corresponding to less than 1.5% and 2% loss from the upper bound, respectively. In comparison, conventional MAB achieves 13.63 dB and 13.59 dB with 30 and 15 codewords, while the PI-based approach reaches only 13.05 dB with 450 codewords and 11.51 dB with 30 codewords. The iterative beamforming method [11] attains 12.31 dB; although convergence is typically reached within 5 iterations, the results here are reported after 10 iterations for consistency. Overall, when using 30 codewords, the proposed method provides a 20% SNR improvement over the PI-based baseline with the same codebook size. Furthermore, even with just 15 codewords, it still outperforms the PI-based search using 450 codewords, reducing the training overhead by a factor of 30 while achieving higher accuracy. Table II shows the average SNR values over 150 trials for different methods under varying levels of UE location error, modeled by σ_{pos} . When $\sigma_{\text{pos}} = 0$, the proposed scheme achieves 13.83 dB,


Fig. 2: Performance comparison of the proposed algorithm with the baseline approaches

which is the highest among all schemes and very close to the hierarchical upper bound in Fig. 2. As the location error increases, all methods experience a performance degradation; however, the proposed scheme consistently maintains the highest SNR. For example, at $\sigma_{\text{pos}} = 3$, it achieves 13.06 dB compared to 12.3 dB for PI-based search with 450 codewords and 12.96 dB for conventional MAB. Even under severe uncertainty with $\sigma_{\text{pos}} = 5$, the proposed scheme yields 11.35 dB, outperforming PI-based with 11.16 dB and MAB with 11.13 dB. These results demonstrate the robustness of the proposed algorithm against location errors, preserving its advantage over existing baselines while requiring significantly fewer training beams.

Table III compares the average number of swept codewords and the runtime per Monte Carlo trial for all methods. All simulations were executed using MATLAB R2024b on a machine equipped with a 12th Gen Intel Core i7-12700H CPU (2.30 GHz, 14 cores) and 16 GB of RAM. Hierarchical search requires sweeping 6400 codewords and takes 1768 s, serving as a near-upper-bound but with prohibitive overhead. In contrast, PI-based, MAB-based, and the proposed method each examine only 30 codewords, yet the proposed approach is fastest at 10.23 s per trial which is 54% faster than PI-based, 35% faster than MAB-based, and 47% faster than iterative beamforming. Overall, the proposed scheme achieves the same low sweep count as the other reduced-search baselines while cutting runtime substantially, and it reduces the number of sweeps by roughly $213\times$ compared to the hierarchical benchmark.

Table IV presents a comparison of the average required transmit power and corresponding energy efficiency for different approaches, assuming a target data rate of 100 Mbps. The proposed scheme achieves near-optimal performance while maintaining low power consumption. Specifically, with only 30 codewords, it requires an average transmit power of

TABLE II: Average SNR (in dB) over 150 trials for different methods under varying standard deviations σ_{pos} of location uncertainty.

σ_{pos}	Iterative Beamforming 10 iteration	PI-based with 450 codewords	PI-based with 30 codewords	MAB with 30 codewords	Proposed with 30 codewords
0	12.26	13.01	11.51	13.59	13.82
3	11.53	12.3	10.96	12.96	13.06
5	9.89	11.16	9.67	11.13	11.35

TABLE III: Average Number of Swept Codewords and Computation Time

Method	Avg. Codewords Swept	Avg. Computation Time per Monte Carlo Trial (s)
Hierarchical Search	6400	1768
PI-Based	30	22.15
MAB-Based	30	15.70
Proposed	30	10.23
Iterative Beamforming [11]	10 iterations	19.22

TABLE IV: Comparison of Beamforming Methods: Required TX Power and Energy Efficiency

Method	Avg. Required TX Power (dBm)	Energy Efficiency (Gbits/J)
Hierarchical	8.13	15.39
Proposed with 30 codewords	8.27	14.89
Proposed with 15 codewords	8.40	14.44
MAB-Based with 30 codewords	8.50	14.12
MAB-Based with 15 codewords	8.55	13.97
PI-Based with 450 codewords	9.08	12.35
Iterative Beamforming [11]	9.82	10.41
PI-Based with 30 codewords	20.33	0.92

8.27 dBm and achieves an energy efficiency of 14.89 Gbits/J, closely approaching the hierarchical benchmark with the results of 8.13 dBm, 15.39 Gbits/J but with significantly lower complexity. Reducing the codebook size to 15 codewords slightly increases the required power to 8.40 dBm while maintaining a competitive energy efficiency of 14.44 Gbits/J. Moreover, the reduced complexity of the proposed scheme also translates into lower computational energy consumption, further contributing to overall energy efficiency in practical deployments. In contrast, conventional PI-based beamforming with 30 codewords performs poorly, requiring 20.33 dBm transmit power and yielding only 0.92 Gbits/J. Even the PI-based with 450 codewords reaches just 12.35 Gbits/J. MAB-based approach offer better efficiency than PI-based baselines but remain inferior to the proposed approach. The iterative beamforming method [11] requires 9.82 dBm and achieves 10.41 Gbits/J, confirming that the proposed method outperforms existing baselines in both energy efficiency and transmit power, even with limited codebook sizes.

VI. CONCLUSION

In this study, an adaptive beamforming strategy for RIS-assisted mmWave communication was presented by integrating PI with a multi-level MAB algorithm to refine beam selection. The proposed approach combines PI-based initialization with online learning and hierarchical refinement, enabling efficient beam management with significantly reduced training complexity. Unlike conventional PI- or MAB-based methods, the algorithm adapts to position errors and maintains robust performance without relying on exhaustive search.

Beyond performance gains, our study highlights the relevance of the proposed scheme for DTs, where virtual models can suggest initial beam choices and real-time feedback ensures continuous adaptation. Such integration links offline planning with online optimization, supporting practical deployment in future 6G networks.

Future work will extend this approach to dynamic mobility and multi-user scenarios, and validate it with real-world positioning systems and hardware platforms. We also plan to address the challenge of generating realistic training data under location uncertainty by leveraging DTs as a flexible platform for training and validation of beamforming algorithms.

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